

WSB HIM&PLC All-in-one User Manual

--V1.79

Applicable to the following models:

4.3 inches:

WS4312R	WS4312T	
WS4320R	WS4320T	
WS4322R	WS4322T	
WSB4322R	WSB4322T	WSB4322MRT
WSB4324T		

7 inches:

WS7020R	WS7020T	
WS7040R	WS7040T	
WS7048R	WS7048T	
WSB7020R	WSB7020T	WSB7020RT
WSB7040R	WSB7040T	WSB7040RT
WSB7048R	WSB7048T	WSB7048RT

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Chapter 1 Product Overview

1.1 Product Overview

Touchscreen section

Download and Communication:

- Program using the Winsunzk HMI Studio 5.1 software. Download the HMI program via USB cable;
- WS series supports USB flash drive downloads, while the WSB series does not support USB flash drive downloads or data export from USB drives.

Product presentation:

- Equipped with a 32-bit ARM9 CPU, it offers fast processing speed.

High-definition LCD screen with a 20,000-hour lifespan; 4-line industrial resistive touchscreen (surface hardness: 4H).

PLC Part:

Download and Communication:

- The download speed is 38.4 Kbps (you may request a reduction to 9.6 kbps under special circumstances). Use Mitsubishi GX Developer or GX Works2 directly for programming, downloading, debugging, and monitoring (monitoring and writing are not supported).

Power supply:

- Powered by a DC 24V supply; under conditions where none of the output relays are activated, the static current is 130 mA; each activated relay increases the current by 12 mA. For example, when all output relays of the 48MR integrated unit are activated, the current reaches 420 mA (10.08 W).

Analog input/output:

- Built-in 6-channel analog input on 7-inch models: 3-channel voltage 0-10V; 3-channel current 0-20mA (4.3-inch models with 20 points and above support 3-channel voltage 0-10V). Can be modified to NTC-10K temperature measurement (range: 0-150°C). 2-channel analog output 0-10V voltage (included in 4.3-inch 22-point models). Model WSB4324T: AD defaults to current mode.
- WSB series offers 4.3-inch and 7-inch models with dual-channel PT100 or K-type thermocouple temperature measurement capability (the 4.3-inch model supports dual K-type/PT100 channels using 3 AD channels and 485 communication interface, while the 7-inch model utilizes 2 AD channels).

WSB series: 4.3-inch and 7-inch models with customizable single-channel PID auto-regulation (supporting K-type, PT100, J-type, and E-type temperature sensors)

Input and output:

- Built-in 6-channel 3K counting, supporting 3-channel AB-phase input (Up to 6-channel 60K

for 7-inch models; up to 2-channel 60K for 4.3-inch models). 4.3-inch MT models (20MT and above): Built-in 2-channel 100K pulse output (expandable to max. 4). Note: On WSB4322RT, COM0 must connect only to 24V negative (Y0/Y1 are transistor outputs, common terminal only for 24V negative). 7-inch MT models: Built-in 4-channel 100K pulse output (expandable to max. 8).

- Relay: 5A rated current, long-term use $\leq 3A$. Transistor: Triode-driven, 1A output current, long-term use $\leq 500mA$.

1.2 Essential parameter

Model	Overall Dimensions L×W (mm)	Mounting Hole Dimensions L×W (mm)	Screen Size	Download Speed	Input Points	Output Points	Output Type	Output Current	Load	High-speed Counting	Pulse Output	Analog Input	Analog Output	MODBUS	Clock	U Disk Download
WS4312R	140*98*36	130*80	4.3"	38.4Kb	8	4	Relay	5A	24V 220V	2/3K	None	None	None	None	None	Support
WS4312T	140*98*36	130*80	4.3"	38.4Kb	8	4	Transistor	1A	24V	2/3K	2/3K	None	None	None	None	Support
WS4320R	140*98*36	130*80	4.3"	38.4Kb	12	8	Relay	5A	24V 220V	2/3K	None	3AD 0-10V	None	Support	Support	Support
WS4320T	140*98*36	130*80	4.3"	38.4Kb	12	8	Transistor	1A	24V	2/3K	2/100K	3AD 0-10V	None	Support	Support	Support
WS4322R	140*98*36	130*80	4.3"	38.4Kb	12	10	Relay	5A	24V 220V	2/60K	None	3AD 0-10V	2DA 0-10V	Support	Support	Support
WS4322T	140*98*36	130*80	4.3"	38.4Kb	12	10	Transistor	1A	24V	2/60K	2/100K	3AD 0-10V	2DA 0-10V	Support	Support	Support
WSB4322R	140*98*36	130*80	4.3"	38.4Kb	12	10	Relay	5A	24V 220V	2/60K	None	Default: 3AD 0-10V / Can be changed to 2PT or 2K temperature measurement	2DA 0-10V	Default supported / Not supported when changed to 2PT 2K	Support	None
WSB4322T	140*98*36	130*80	4.3"	38.4Kb	12	10	Transistor	1A	24V	2/60K	2/100K		2DA 0-10V		Support	None
WSB4322RT	140*98*36	130*80	4.3"	38.4Kb	12	10	2-channel Transistor, 8-channel Relay	1A	24V 220V	2/60K	2/100K	Default: 3AD 0-20mA / Can be changed to 2PT or 2K temperature measurement	2DA 0-10V		Support	None
WSB4324T	140*98*36	130*80	4.3"	38.4Kb	12	12	Transistor	1A	24V	2/60K	2/100K		2DA 0-10V		Support	None
WS7020R	199*144*59	190*136	7"	38.4Kb	12	8	Relay	5A	24V 220V	6/3K	None	3AD 0-10V 3AD 0-20mA	2DA 0-10V	Support	Support	Support
WS7020T	199*144*59	190*136	7"	38.4Kb	12	8	Transistor	1A	24V	6/3K	4/100K	3AD 0-10V 3AD 0-20mA	2DA 0-10V	Support	Support	Support
WS7040R	199*144*59	190*136	7"	38.4Kb	24	16	Relay	5A	24V 220V	6/3K	None	3AD 0-10V 3AD 0-20mA	2DA 0-10V	Support	Support	Support
WS7040T	199*144*59	190*136	7"	38.4Kb	24	16	Transistor	1A	24V	6/3K	4/100K	3AD 0-10V 3AD 0-20mA	2DA 0-10V	Support	Support	Support
WS7048R	199*144*59	190*136	7"	38.4Kb	24	24	Relay	5A	24V 220V	6/3K	None	3AD 0-10V 3AD 0-20mA	2DA 0-10V	Support	Support	Support
WS7048T	199*144*59	190*136	7"	38.4Kb	24	24	Transistor	1A	24V	6/3K	4/100K	3AD 0-10V 3AD 0-20mA	2DA 0-10V	Support	Support	Support
WS7B7020R	202*148*51	190*136	7"	38.4Kb	12	8	Relay	5A	24V 220V	6/3K	None	Default: 3AD 0-10V, 3AD 0-20mA / AD0, AD1 can be changed to 2PT or 2K temperature measurement	2DA 0-10V	Support	Support	None
WS7B7020T	202*148*51	190*136	7"	38.4Kb	12	8	Transistor	1A	24V	6/3K	4/100K		2DA 0-10V	Support	Support	None
WS7B7020RT	202*148*51	190*136	7"	38.4Kb	12	8	4-channel Transistor, 4-channel Relay	1A	24V 220V	6/3K	4/100K		2DA 0-10V	Support	Support	None
WS7B7040R	202*148*51	190*136	7"	38.4Kb	24	16	Relay	5A	24V 220V	6/3K	None		2DA 0-10V	Support	Support	None
WS7B7040T	202*148*51	190*136	7"	38.4Kb	24	16	Transistor	1A	24V	6/3K	4/100K		2DA 0-10V	Support	Support	None
WS7B7040RT	202*148*51	190*136	7"	38.4Kb	24	16	4-channel Transistor, 12-channel Relay	1A	24V 220V	6/3K	4/100K		2DA 0-10V	Support	Support	None
WS7B7048R	202*148*51	190*136	7"	38.4Kb	24	24	Relay	5A	24V 220V	6/3K	None		2DA 0-10V	Support	Support	None
WS7B7048T	202*148*51	190*136	7"	38.4Kb	24	24	Transistor	1A	24V	6/3K	4/100K		2DA 0-10V	Support	Support	None
WS7B7048RT	202*148*51	190*136	7"	38.4Kb	24	24	4-channel Transistor, 20-channel Relay	1A	24V 220V	6/3K	4/100K		2DA 0-10V	Support	Support	None

Note: Some functions of the Winsunzk all-in-one HMI can be customized; please contact customer service for consultation.

1. The WSB series all-in-one machine is optional with 2-channel PT100 and 2-channel K-type thermocouple, and optional PID auto-tuning temperature measurement (PID supports K-type, PT100, J-type, and E-type temperature measurement).

When the 4.3-inch WSB model is equipped with temperature measurement function, the 485 port will be occupied; when equipped with 2PT and 2K temperature measurement, the 485 function is unavailable. The AD input of model WSB4324T defaults to current mode.

2. The WSB series does not support USB download and USB data export functions, while the WS series supports them.

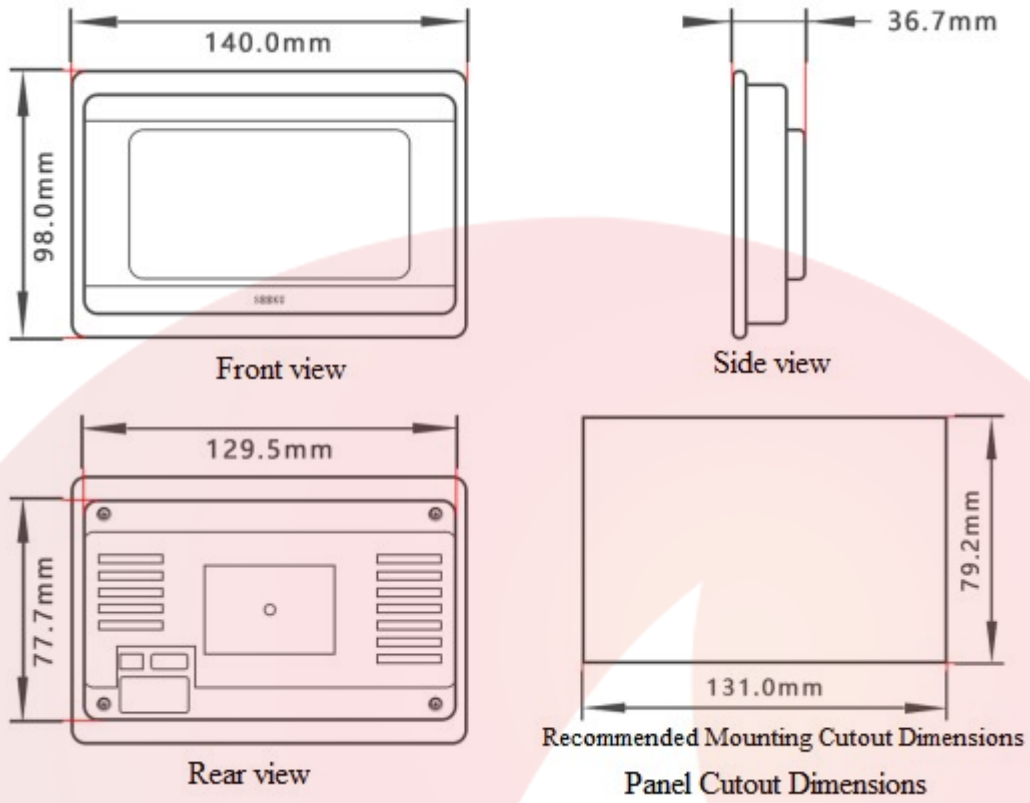
3. The 7-inch all-in-one machine supports a maximum of 6 channels of 60K high-speed counting and a maximum of 8 channels of 100K high-speed pulse output.

4. Analog input can be modified to NTC-10K temperature measurement (0-150°C) or current (0-20mA).

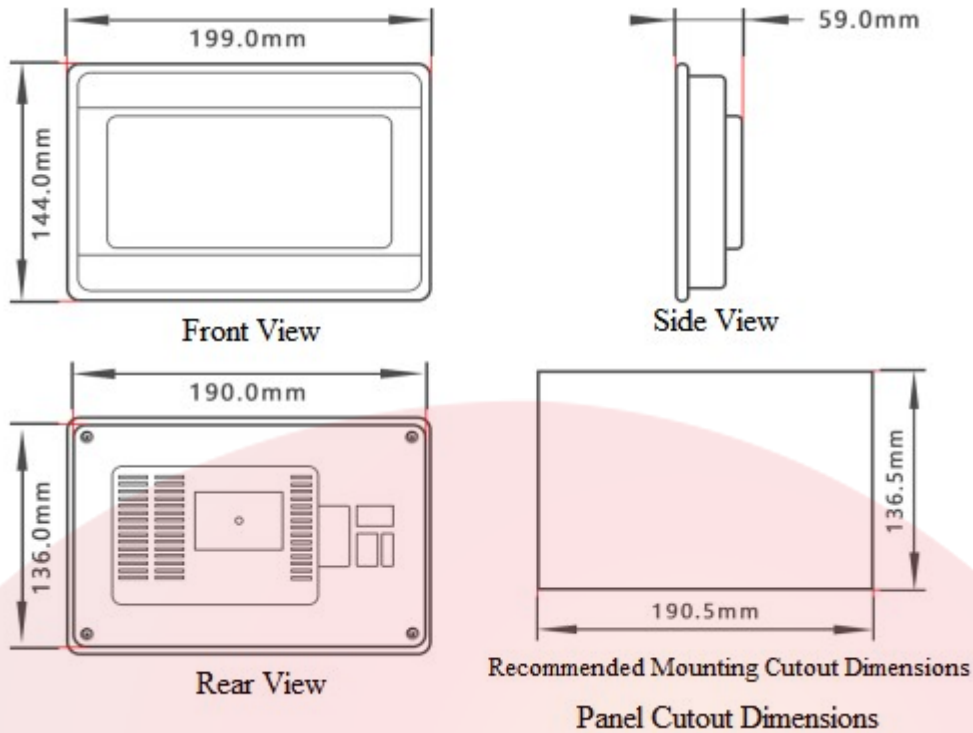
1.3 External dimensions and installation method

1.3.1 External dimensions

4.3-inch all-in-one computer installation dimension diagram



7-inch all-in-one machine installation dimension diagram



1.3.2 Method of installation

The installation schematic is as follows:

⚠ Torque must not exceed 0.2 N·m (Figure 2)

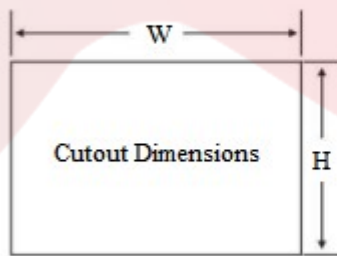


Figure 1

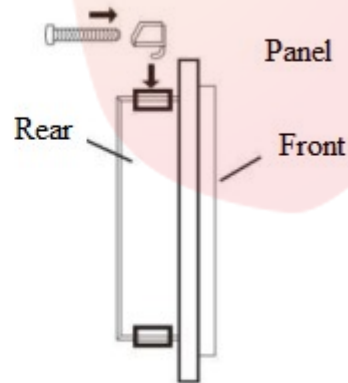


Figure 2

Specific installation steps:

Install the product on the control panel of the control cabinet. Use a Phillips-head screwdriver and the included mounting hardware for installation. Follow the steps below to complete the installation.

1. On the panel surface, cut a rectangular mounting groove and drill the embedding holes according to the specified dimensions, then install the HMI unit from the front side. (Figure 1)

2. Insert the panel mounting brackets (included) into the four mounting holes at the bottom of the rear panel of the HMI unit, then insert the mounting screws (included) and tighten them with a Phillips-head screwdriver.

Chapter 2 Product Display

2.1 Main Product Hardware Specifications

WS 4.3-inch all-in-one computer

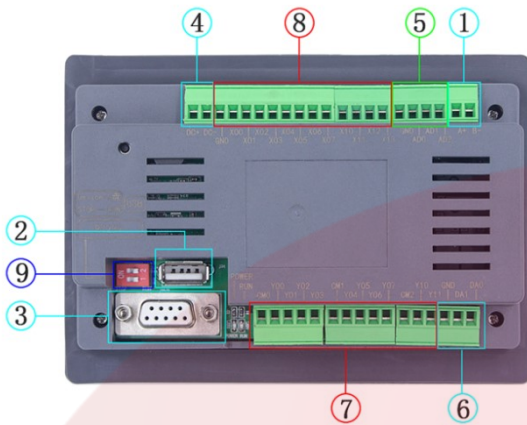


Figure 2-1

WS 7-inch all-in-one computer

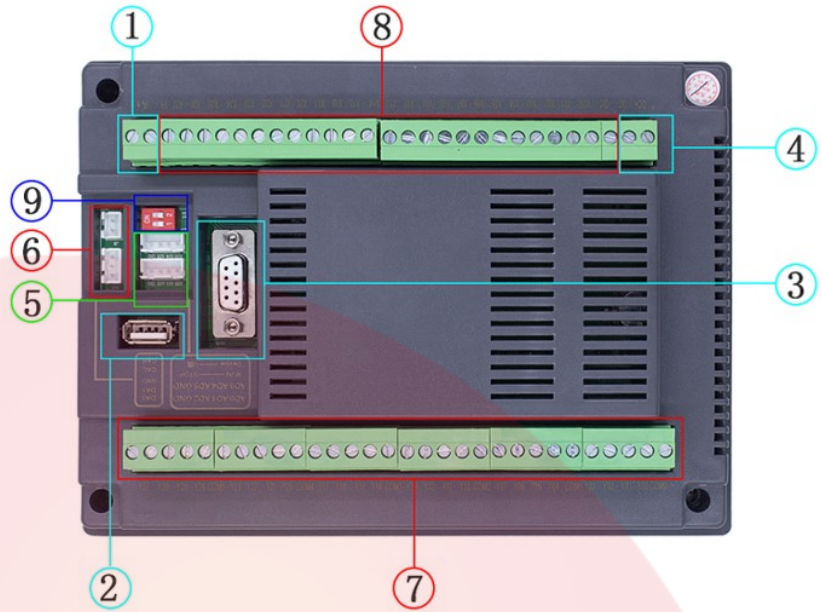


Figure 2-2

WSB 4.3-inch all-in-one computer

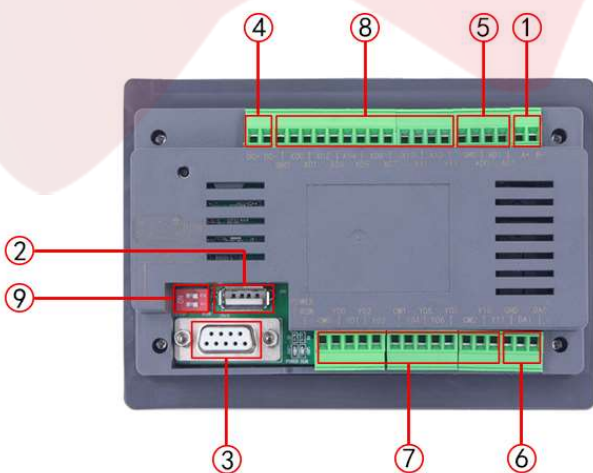


Figure 2-3

WSB 7-inch all-in-one computer

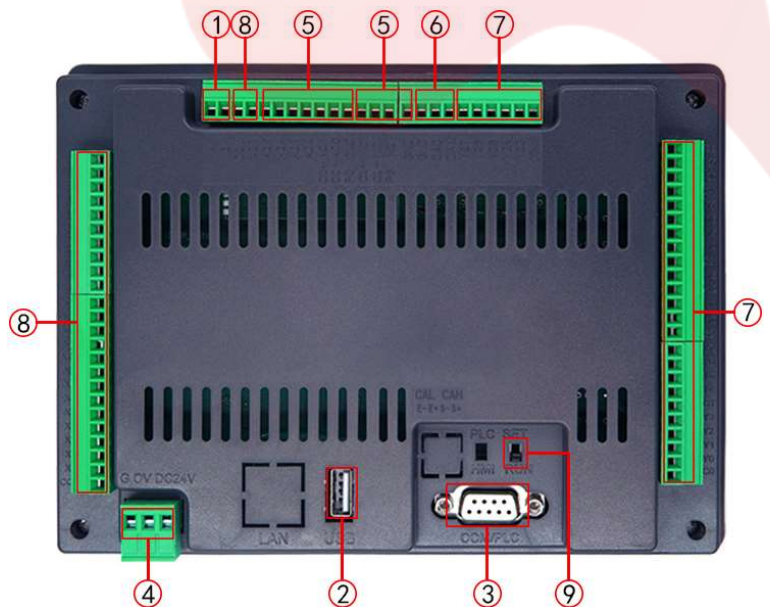


Figure 2-4

Port Description:

- ①RS485 Interface
- ②Touch screen download port: supports USB A-to-A cable download and U-disk download; WSB series does not support U-disk.
- ③PLC Download Port
- ④DC24V Power Terminal
- ⑤Analog Input
- ⑥Analog Output
- ⑦Output Terminal
- ⑧Input Terminal
- ⑨DIP Switch: The first gear is PLC switch. Toggle to RUN (right) for running state, toggle to STOP (left) for stop state.

2.2 Product Rendering

WS 4.3-inch front view



Figure2-5

WS 4.3-inch back view image



Figure2-6

WS 7-inch front view

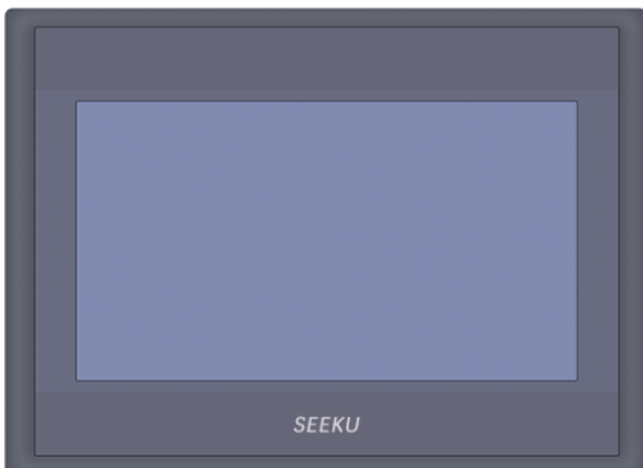


Figure2-7

WS 7-inch back view image



Figure2-8

WSB 4.3-inch front view



Figure2-9

WSB 4.3-inch back view



Figure2-10

WSB 7-inch front view



Figure 2-11

WSB 7-inch back view image

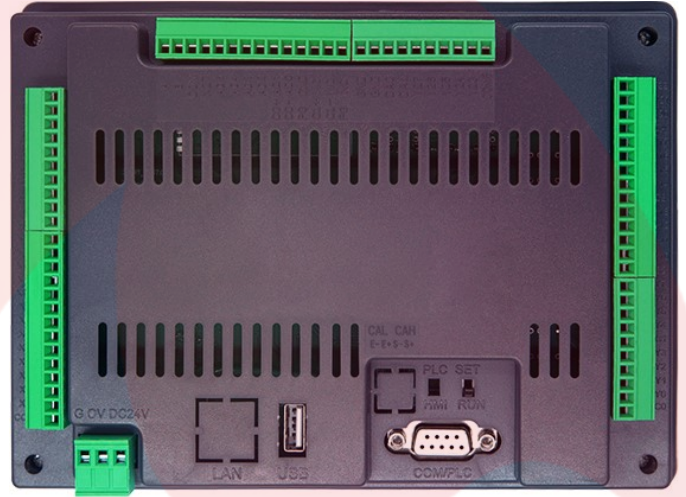


Figure2-12

Chapter 3 HMI Parameters and Programming Instructions

3.1 Basic parameters of the HMI

	Model	WS-043AO (WS43 series)	WS-070AO(WS70 series)
Software performance	Widget Image Memory	Approx. 45MB	Approx. 45MB
	Screen Image Memory	Approx. 50MB	Approx. 50MB
	Power-off Address Retention	Supports power-off retention	Supports power-off retention
	Historical Alarms	Supports saving 50 records at a time (optional power-off retention available)	Supports saving 50 records at a time (optional power-off retention available)
	Historical Data	Supports saving 50 records at a time (optional power-off retention available)	Supports saving 50 records at a time (optional power-off retention available)
	Max. Widgets per Screen	255	255
	Current Page Memory	6.7M	6.7M
	Historical Sampling Memory	63K	63K
Hardware performance	Display	4.3" TFT LCD Display	7" TFT LCD Display
	Resolution	480×272	800X480
	Color	260K Colors	260K Colors
	Brightness	400 cd/m ²	400 cd/m ²
	Backlight	LED	LED
	LED Lifespan	20,000 hours	20,000 hours
	Touch Screen	4-wire industrial resistive touch screen (surface hardness: 4H)	4-wire industrial resistive touch screen (surface hardness: 4H)
	CPU	32-bit 240MHz ARM9	32-bit 240MHz ARM9
	Memory	128M NAND Flash	128M FLASH
	RTC	Built-in real-time clock	Built-in real-time clock
	Ethernet	None	None
	USB Port	1× USB2.0 Device / 1× USB2.0 Host port	1× USB2.0 Device / 1× USB2.0 Host port

	Program Download Method	USB download	USB download
	USB Disk	Supports USB disk upgrade, data import & export	Supports USB disk upgrade, data import & export
Electrical specifications	Rated Power	<5W	<10W
	Rated Voltage	DC 24V, operating range $\pm 15\%$	DC 24V, operating range $\pm 15\%$
	Power Protection	Built-in lightning and surge protection	Built-in lightning and surge protection
	Allowable Power Loss	< 5ms	< 5ms
	CE&RoHS	Compliant with EN61000-6-2:2005 & EN61000-6-4:2007; Lightning/surge $\pm 4\text{kV}$, burst $\pm 4\text{kV}$; ESD contact $\pm 4\text{kV}$, air discharge $\pm 8\text{kV}$	Compliant with EN61000-6-2:2005 & EN61000-6-4:2007; Lightning/surge $\pm 4\text{kV}$, burst $\pm 4\text{kV}$; ESD contact $\pm 4\text{kV}$, air discharge $\pm 8\text{kV}$
Environmental specification	Operating Temperature	-20 ~ 50°C	-20 ~ 50°C
	Storage Temperature	-40 ~ 70°C	-40 ~ 70°C
	Ambient Humidity	10 ~ 90% RH (non-condensing)	10 ~ 90% RH (non-condensing)
	Vibration Resistance	10 ~ 25 Hz (X/Y/Z axes, 2G/30 min)	10 ~ 25 Hz (X/Y/Z axes, 2G/30 min)
	Cooling-down method	Natural air cooling	Natural air cooling

	Model	WS-043DP (The WSB series offers an optional 2K 2PT temperature measurement feature.)	WS-070DP (The WSB series offers an optional 2K 2PT temperature measurement feature.)
Software performance	Widget Image Memory	Approx. 18MB	Approx. 18MB
	Screen Image Memory	Approx. 17MB	Approx. 17MB
	Power-off Address Retention	Supports 3-second detection and saving	Supports 3-second detection and saving
	Historical Alarms	Supports saving 50 items at once	Supports saving 50 items at once
	Historical Data	Supports saving 50 items at once	Supports saving 50 items at once

	Max. Widgets per Screen	100	100
	Current Page Memory	1.5M	1.5M
	Historical Sampling Memory	63K	63K
Hardware performance	Display	4.3" TFT LCD Display	7" TFT LCD Display
	Resolution	480×272	800×480
	Color	65536	65536
	Brightness	250 cd/m2	250 cd/m2
	Backlight	LED	LED
	LED Lifespan	20000 hours	20000 hours
	Touch Screen	4-line industrial resistance touchscreen (surface hardness: 4H)	4-line industrial resistance touchscreen (surface hardness: 4H)
	CPU	600MHz ARM9 with built-in 8MB DDR2 memory	600MHz ARM9 with built-in 8MB DDR2 memory
	Memory	16Mbyte SPI NORFlash	16Mbyte SPI NORFlash
	RTC	Built-in real-time clock	Built-in real-time clock
	Ethernet	None	None
	USB Port	1 × USB2.0 Device Port / 1 × USB2.0 Host Port	1 × USB2.0 Device Port / 1 × USB2.0 Host Port
	Program Download Method	USB download	USB download
	USB Disk	Nonsupport	Nonsupport
Electrical specifications	Rated Power	< 5W	< 5W
	Rated Voltage	DC 24V, operating range ±15%	DC 24V, operating range ±15%
	Power Protection	Features lightning and surge protection	Features lightning and surge protection
	Allowable Power Loss	< 5 mS	< 5 mS
	CE&RoHS	Compliant with EN61000-6-2:2005 and EN61000-6-4:2007 standards; Lightning	Compliant with EN61000-6-2:2005 and EN61000-6-4:2007 standards; Lightning

		surge $\pm 250\text{V}$, electrical fast transient $\pm 2\text{KV}$; contact electrostatic discharge $\pm 4\text{KV}$, air electrostatic discharge $\pm 8\text{KV}$.	surge $\pm 250\text{V}$, electrical fast transient $\pm 2\text{KV}$; contact electrostatic discharge $\pm 4\text{KV}$, air electrostatic discharge $\pm 8\text{KV}$.
Environmental specification	Operating Temperature	$-10\sim 50^{\circ}\text{C}$	$-10\sim 50^{\circ}\text{C}$
	Storage Temperature	$-30\sim 70^{\circ}\text{C}$	$-30\sim 70^{\circ}\text{C}$
	Ambient Humidity	10~90%RH (No condensation)	10~90%RH (No condensation)
	Vibration Resistance	10 ~ 25 Hz (2G for X, Y, Z axis / 30 minutes)	10 ~ 25 Hz (2G in X, Y, Z directions, 30 minutes)
	Cooling-down Method	Natural air cooling	Natural air cooling

3.2 Touchscreen Programming Instructions

3.2.1 The touchscreen employs the Weisheng HMI 5.1 software, while the WSB touchscreen uses the Weisheng 5.1 version for programming, with data downloaded via a USB male-to-male cable.

3.2.2 The software includes detailed user manuals for operating and using the programming software.



3.3 Touchscreen and PLC Communication Specifications

3.3.1 The PLC does not require communication parameter configuration; communication settings are handled within the touchscreen software.

3.3.2 The touchscreen uses the Mitsubishi FX1N protocol, with a baud rate of 38400 and a communication mode of 232.

3.3.3 After the program download is complete, disconnect the download cable; the PLC and touchscreen will automatically communicate. (Note: When the download cable is connected, the PLC and touchscreen cannot communicate.)



The image displays two screenshots of the touchscreen software interface. The top screenshot shows the '协议' (Protocol) dialog box, which is used to configure the communication protocol. It includes fields for '制造商' (Manufacturer) set to 'MITSUBISHI (三菱)', '系列' (Series) set to 'Mitsubishi_Fx1n', and '名称' (Name) set to 'Mitsubishi_Fx1n'. The bottom screenshot shows the '连接设置' (Connection Settings) dialog box. It is divided into two sections: '连接属性' (Connection Properties) and '协议属性' (Protocol Properties). In the '连接属性' section, '连接名称' (Connection Name) is 'COM1' and '连接端口' (Connection Port) is 'COM1'. In the '协议属性' section, '波特率' (Baud Rate) is '38400', '停止位' (Stop Bits) is '1', '校验' (Parity) is 'EVEN', '数据位' (Data Bits) is '7', and '通信方式' (Communication Mode) is '232'. At the bottom of the dialog box, there are four buttons: '恢复默认设置' (Restore Default Settings), '确定' (OK), '取消' (Cancel), and '帮助' (Help).

Chapter 4 PLC Electrical Design Reference

4.1 Power Supply and Power Consumption

Powered by DC 24;

Under the condition that none of the output relays of the 7-inch all-in-one machine are activated, the static current is 190 mA;

For each channel activated, the current increases by 12.5 mA. For example, when the WS7048R output relay is fully activated, the current reaches 490 mA (11.76 W);

Note: Select a switching power supply with minimal ripple for power delivery. When the circuit is subject to strong interference, use an appropriate filter for noise reduction.

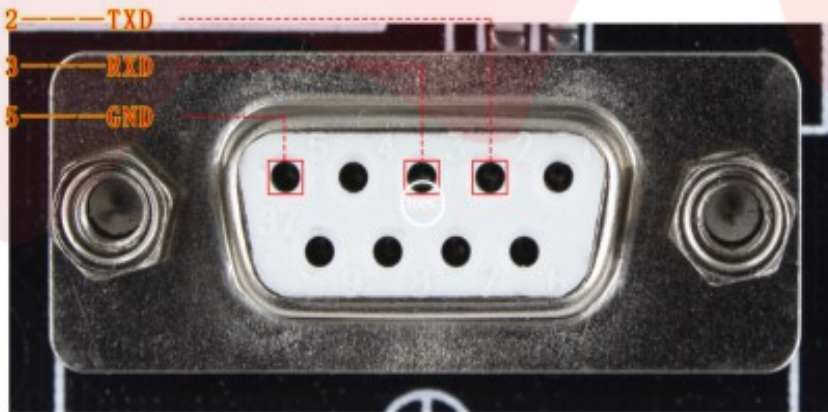
4.2 232 Communication Port Description

- By default, it comes with a set of 232 interfaces, which are used for program download. You can use a USB-to-serial cable or a male-to-female direct cable to download the program.

Communication Interface Definition

- 2——TXD (Send data)
- 3——RXD (Receive data)
- 5——GND (Signal Ground)

PLC-side RS232 9-pin port



4.3 485 Communication Port Description

Serial data transmission:

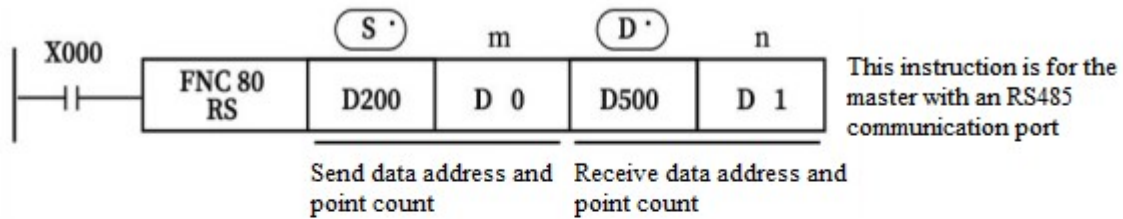
Special Register	Instruction	Special Relay	Instruction
RS485 communication port			
D8120	Definition of RS485 Communication Format	M8121	Set when sending data and reset automatically after completion
D8121	RS485 Communication Station Number Configuration	M8122	Send a request: When the M8122 is activated, data transmission begins as soon as an idle communication port becomes available, and the system automatically resets upon initiation of transmission.
D8122	Remaining data for transmission	M8123	The data reception completion flag is automatically set upon receiving a frame of data. Users should reset this flag after data reception.
		M8124	Set during data reception, reset after data reception
M8129: Communication timeout flag. When the master sends a command and the slave fails to respond within the D8129 time limit, M8129 is set.			

The corresponding communication parameters for D8120 and D8126 are as follows:

Item	Name	Content	
		0 (bit OFF)	1 (bit ON)
B0	Data length	7-bit	8-bit
B1 B2	parity bit	b2 b1 (0,0) No parity (0,1): ODD (1,1): Even parity EVEN	
B3	stop bit	1-bit	2-bit
B4 B5 B6 B7	transfer rate bps	b7 b6 b5 b4 b7 b6 b5 b4 {0, 0, 1, 1}: 300 {0, 1, 1, 1}: 4800 {0, 1, 0, 0}: 600 {1, 0, 0, 0}: 9600 {0, 1, 0, 1}: 1200 {1, 0, 0, 1}: 19200 {0, 1, 1, 0}: 2400 {1, 0, 1, 0}: 38400	
B8	start symbol	None	Have (D8124)
B9	Terminal symbol	None	Have (D8125)
B10 B11		Do not use	
B12 B13 B14 B15	communication protocol	B15 b14 b13 b12 {0, 0, 0, 0}: Mitsubishi FX2N protocol (Slave) {0, 1, 0, 0}: MODBUS RTU (Slave) {1, 0, 0, 0}: MODBUS RTU (Master, IVRD, IVWR commands) {1, 1, 0, 0}: Free communication (RS command, using CCD parity)	

When the M8120 is reset and RS is executed, the parameters provided are for the RS485 port; when

the M8120 is set and RS is executed, the parameters are for the RS232 port.



- The data transmission format can be configured using the special data register D8120, as described below. However, changes to the D8120 settings are not actually applied while the RS instruction is active.
- In systems that do not transmit data, set the number of data transmission points to "K0." Similarly, in systems that do not receive data, set the number of reception points to "K0."

Figure 4-1

CCD instruct:

Stores the sum of each bit of n-point data starting from the element specified by S, and the CRC check data into D, D+2, and D+3. In this example, the checksum is stored in D0, and the CRC check data is stored in D2 and D3.

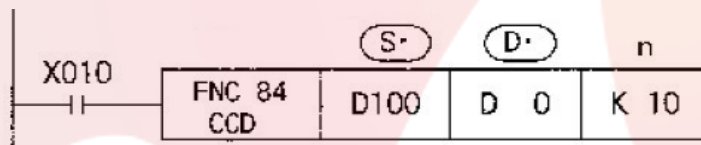


Figure 4-2

4.4 Communicating with frequency converters or instruments:

Read:

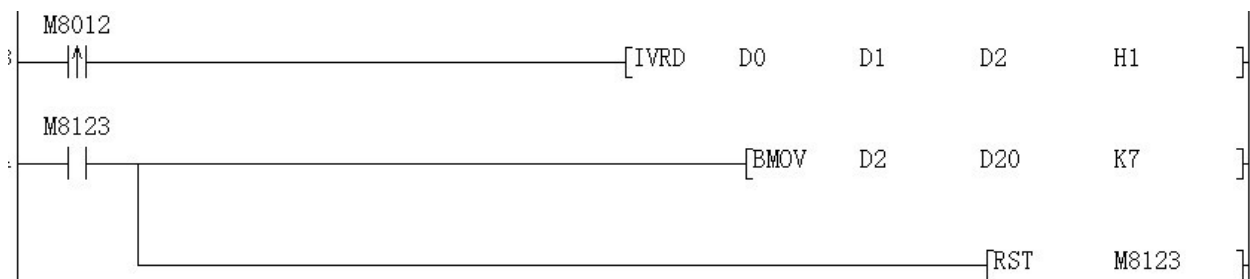
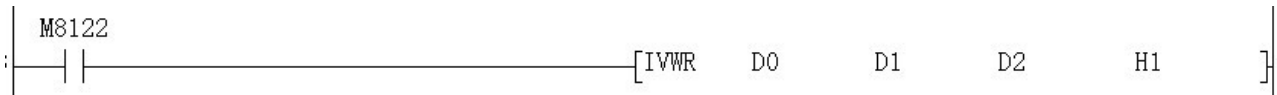


Figure 4-3

D0 stores the read station number (high 8 bits) and command code (low 8 bits). If D0 = H103, it means station number 1 and read command 3. D1 is the address of the data to be read. D2 is the starting

address for data returned by the frequency converter or instrument. When data is received on channel 0, M8123 will be set. H1: high 8 bits = channel number, low 8 bits = number of data to read. Reading 1 piece of data via channel 0 (RS485 channel). If the value is H101, it means reading 1 piece of data via channel 1 (RS232 channel).

Write in:

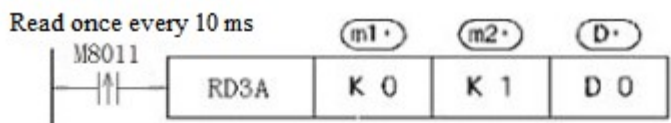


D0 stores the station number (high 8 bits) and command code (low 8 bits) for writing. For example, if the value of D0 is H106, it means station number 1 and single data write command 6. D1 is the address of the data to be written. D2 is the starting address of the data to be written to the frequency converter or instrument. For H1: high 8 bits represent the channel number, and low 8 bits represent the number of data to be written. Writing 1 piece of data via channel 0 (RS485 channel). If the value is H101, it means writing 1 piece of data via channel 1 (RS232 channel). M8122 will be automatically set upon completion of writing.

4.5 Analog Input/Output Description

4.5.1 Analog Reading Instruction:

AD0, AD1, and AD2 are 0 – 10 V analog inputs, while AD3, AD4, and AD5 are 0 – 20 mA analog inputs. The corresponding values for 0 – 10 V and 0 – 20 mA range from 0 to 4095.



● Command for reading analog input values from an analog module.

m1: Module number; set to K0 on the host

m2: Analog input channel number
K0–K5 (corresponding to AI1–6)

D: Save the instantaneous data value to D0
Saves the value read from the analog module.

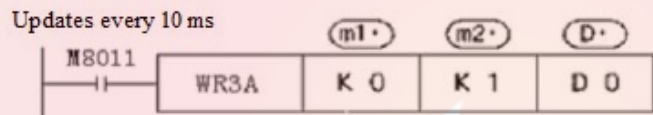
Figure 4-5



Figure 4-6 Example of an analog input program

4.5.2 Analog Output Instruction:

DA0 and DA1 provide 0 – 10 V analog outputs, corresponding to values ranging from 0 to 4095.



● Command for writing digital values to an analog module

m1: Module number; set to 0 on the host

m2: Analog output channel number K0-K1

D: Data to be written; specifies the value to be written to the analog module (0-4095)

Figure 4-7

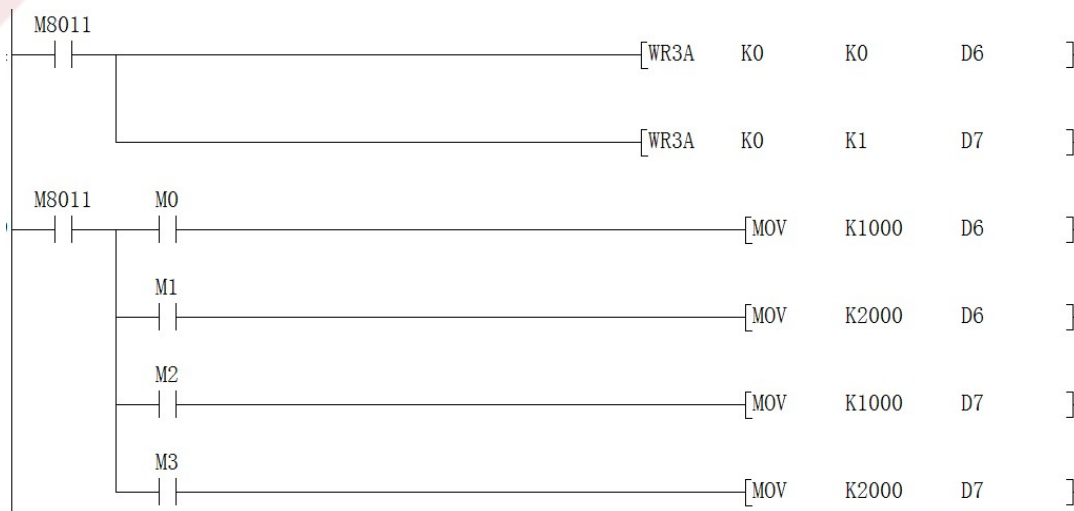


Figure 4-8 Example of an analog output program

4.6 Clock Module Description:

When setting the clock, M8015 should be set. When resuming operation, M8015 should be reset. D8018: Year, D8017: Month, D8016: Day, D8019: Week, D8015: Hour, D8014: Minute, D8013: Second. The clock data read instruction TRD can be used to read clock data into general registers, or the clock write instruction TWR can be used to modify the clock. Using these instructions does not require setting M8015.

4.7 PID Operation Instruction Description:

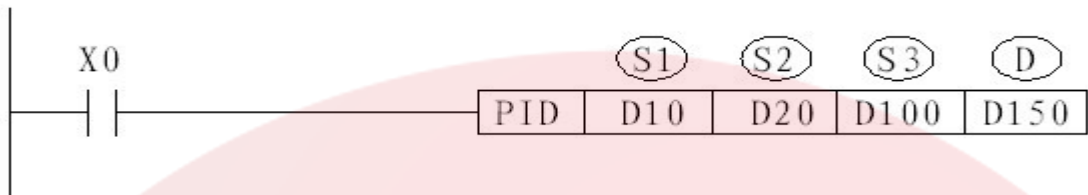


Figure 4-9

This instruction is used for PID operation programs for PID control. S1: Set target value; S2: Current value (feedback value); S3: PID control parameters, occupying 9 consecutive D registers starting from S3. S3: PID channel number; S3+1: Proportional coefficient KP; S3+2: Integral coefficient KI; S3+3: Derivative coefficient KD; S3+4: Error coefficient KE, PID processing is performed only when the error exceeds this value; S3+5: Output upper limit value PMAX; S3+6: Output lower limit value PMIN; S3+7: Reserved; S3+8: Reserved; D: Control value output.

4.8 High-speed Count:

The SPD instruction (supporting X0-5): If an encoder completes 360 pulses per revolution, a 2x frequency multiplication yields 720 pulses, while a 4x frequency multiplication produces 1440 pulses, thereby enhancing the encoder's resolution.

Countinput	Single-phase counter number	Up/Down Counting Direction Switch	Countinput	Single-phase 2-frequency counter	Up/Down Counting Direction Switch
X0	C235	M8235	X0	C241	M8241
X1	C236	M8236	X1	C242	M8242
X2	C237	M8237	X2	C243	M8243
X3	C238	M8238	X3	C244	M8244
X4	C239	M8239	X4	C245	M8245
X5	C240	M8240	X5	C246	M8246

Countinput	Bipolar 2x frequency multiplier counter number	Up and down counting direction (read-only)	Countinput	Bipolar 4x frequency multiplier counter number	Up and down counting direction (read-only)
X0 (Phase A)	C250	M8250	X0 (Phase A)	C253	M8253
X1 (Phase B)			X1 (Phase B)		

X2 (Phase A)	C251	M8251	X2 (Phase A)	C254	M8254
X3 (Phase B)			X3 (Phase B)		
X4 (Phase A)	C252	M8252	X4 (Phase A)	C255	M8255
X5 (Phase B)			X5 (Phase B)		

C247 (X0, X1), C248 (X2, X3), and C249 (X6, X7) are non-duplexed bi-phase counters.

4.9 High-speed pulse output and pulse width modulation:

Supports 8-channel pulse outputs Y0 – Y7 (PLSY, PLSV, PLSR, DRVA, DRVI) or 6-channel pulse width modulation (PWM) outputs Y0 – Y5, operating at a frequency of 100 kHz.

Pulse	Output pulse count	Output tag	Pulse inhibit	Minimum output frequency	Acceleration and deceleration	DSZR DVIT Direction	DVIT interrupt input X address	Origin Regression Speed	Original Origin Regression on Walking	ZRN crawling pulse count
Y0	D8132	M8147	M8141	D8144	D8145	M8080	D8080	D8220	D8090	D8072
Y1	D8134	M8148	M8142	D8146	D8147	M8081	D8081	D8221	D8091	D8073
Y2	D8136	M8149	M8143	D8148	D8149	M8082	D8082	D8222	D8092	D8074
Y3	D8138	M8150	M8144	D8150	D8151	M8083	D8083	D8223	D8093	D8075
Y4	D8140	M8151	M8145	D8152	D8153	M8084	D8084	D8224	D8094	D8076
Y5	D8142	M8152	M8146	D8154	D8155	M8085	D8085	D8225	D8095	D8077
Y6	D8166	M8153	M8155	D8156	D8157	M8086	D8086	D8226	D8096	D8078
Y7	D8168	M8154	M8156	D8158	D8159	M8087	D8087	D8227	D8097	D8079

4.10 Interrupt Description:

1.External interrupts support X0-X5, with interrupt numbers listed in the table below:

	Rising edge	Trailing edge	Interrupt inhibit
X0	I0	I1	M8050
X1	I100	I101	M8051
X2	I200	I201	M8052
X3	I300	I301	M8053
X4	I400	I401	M8054
X5	I500	I501	M8055

2.The timer interrupt pointer is I600, and the interrupt disable is M8056. The interrupt time range is I601 (1 ms) to I699 (99 ms).

3.Counter Interrupt Pointer

Pointer number	Interrupt inhibit
I10	M8059
I20	
I30	
I40	
I50	
I60	

4.11 Product wiring diagram

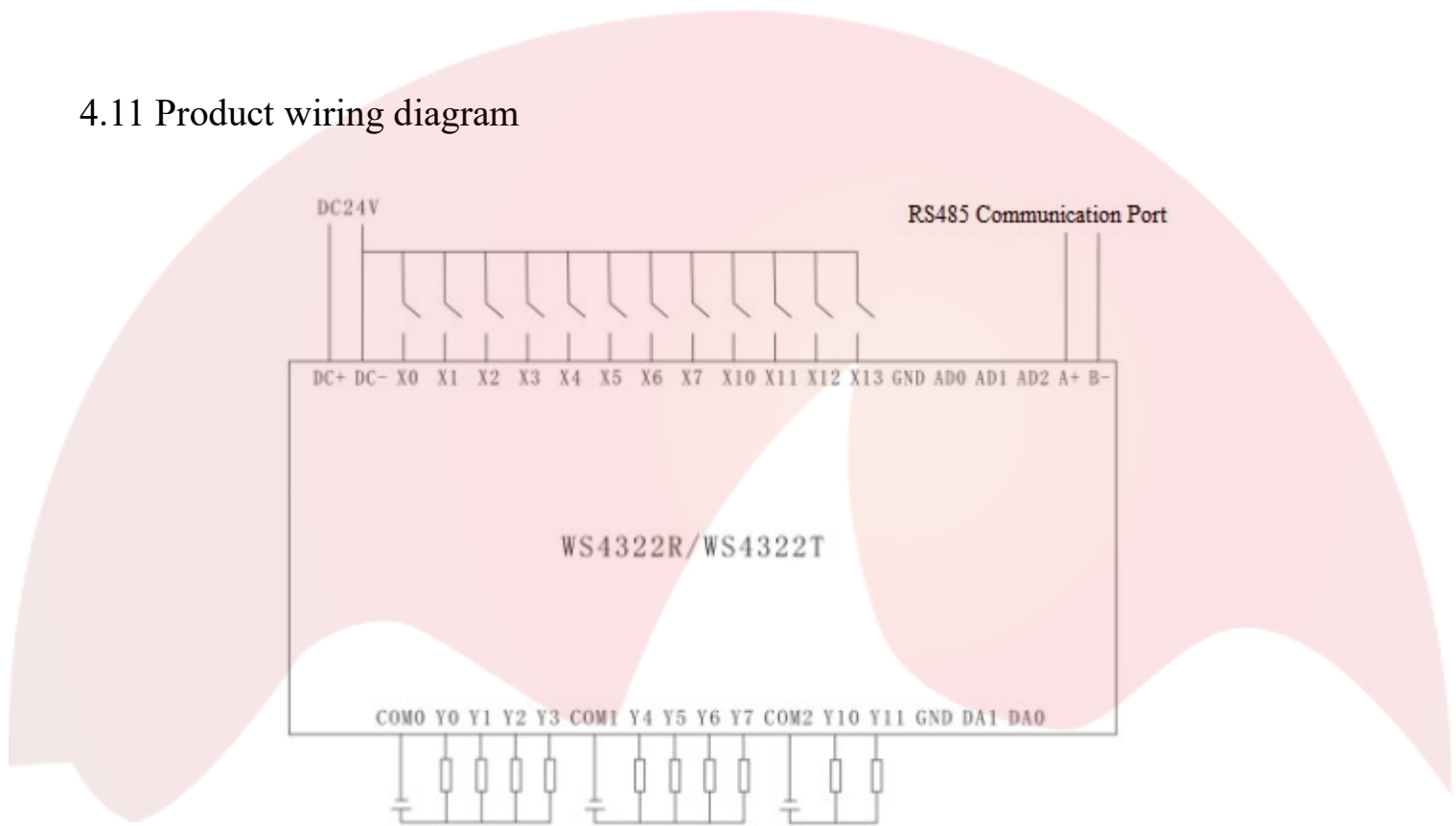


Figure 4-10

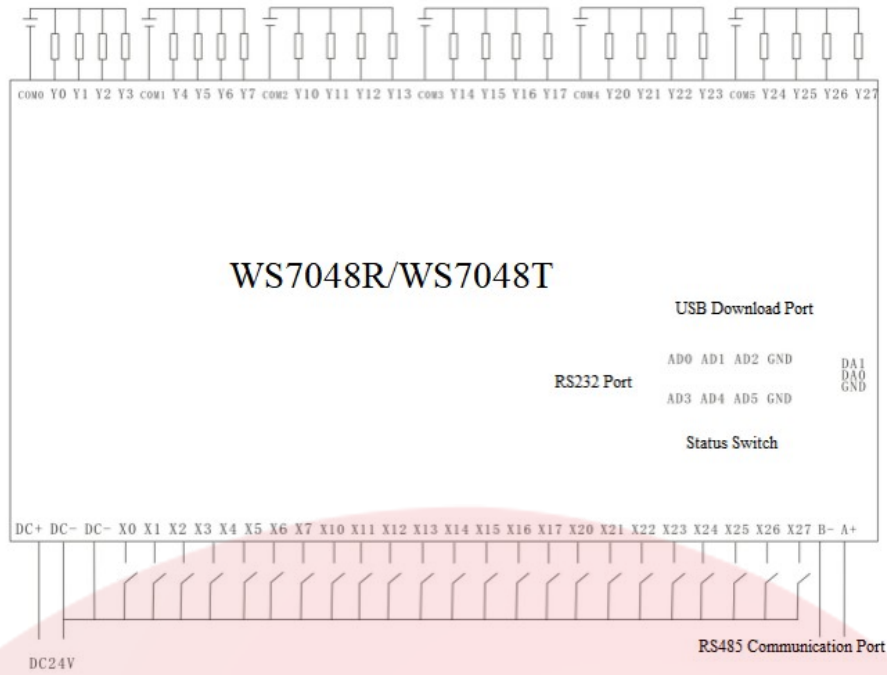


Figure 4-11

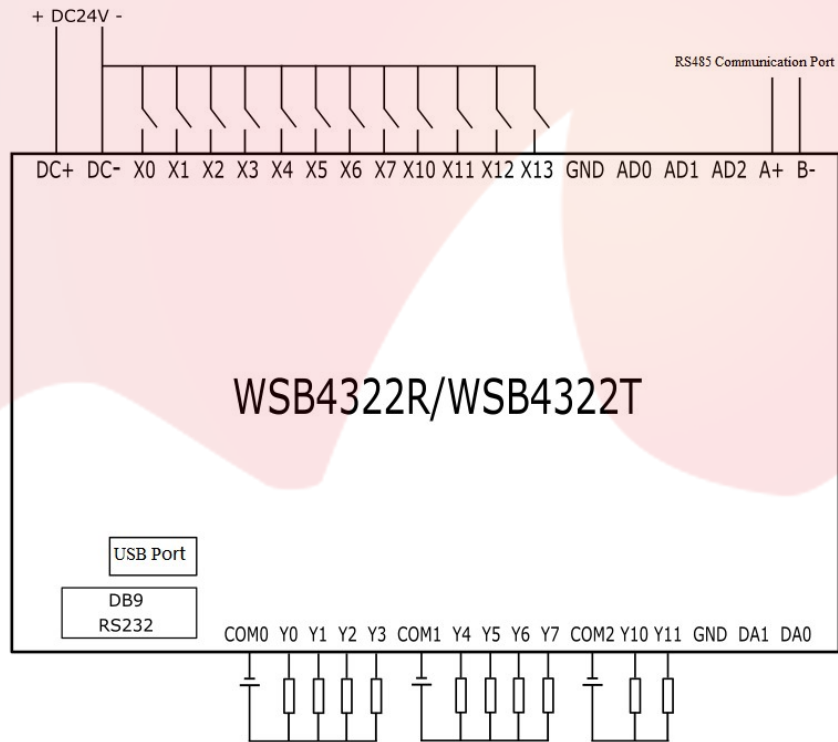


Figure 4-12

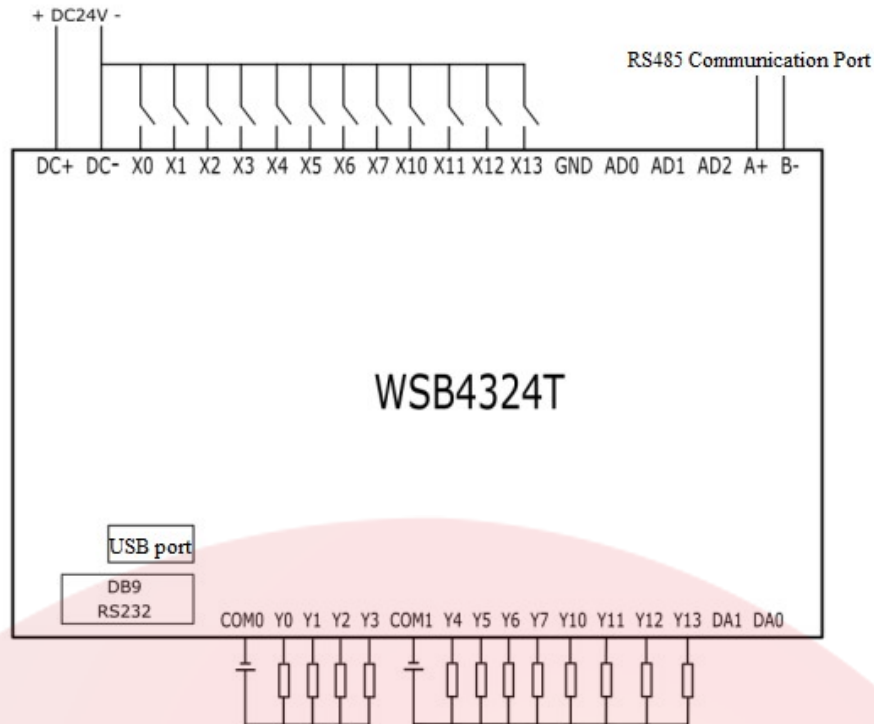


Figure 4-13

When selecting the PT100 K-type thermocouple for temperature measurement, the wiring connection to the AD terminal is shown in the figure below.

Applicable to (WSB4322R/WSB4322T)

GND corresponds to P0+ K0+, AD0 corresponds to P0- K0-, AD1 corresponds to PG NC.

AD2 corresponds to P1+ K1+, A+ corresponds to P1- K1-, B- corresponds to PG NC.

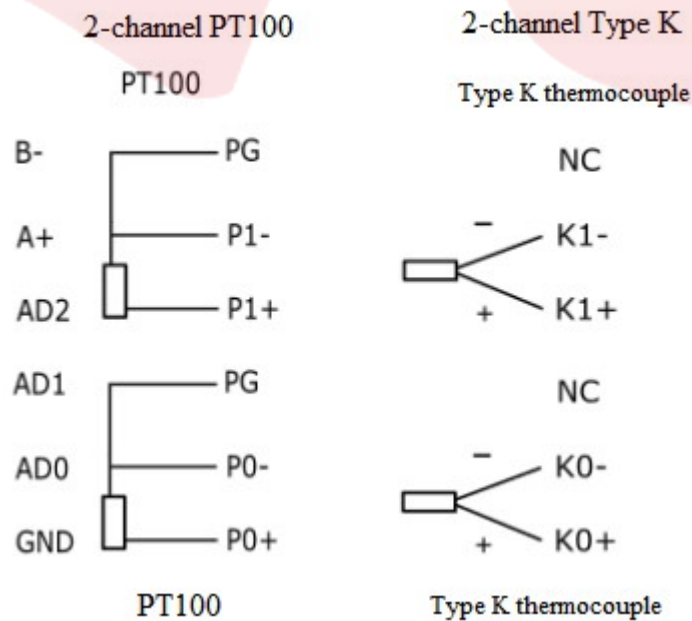


Figure 4-14

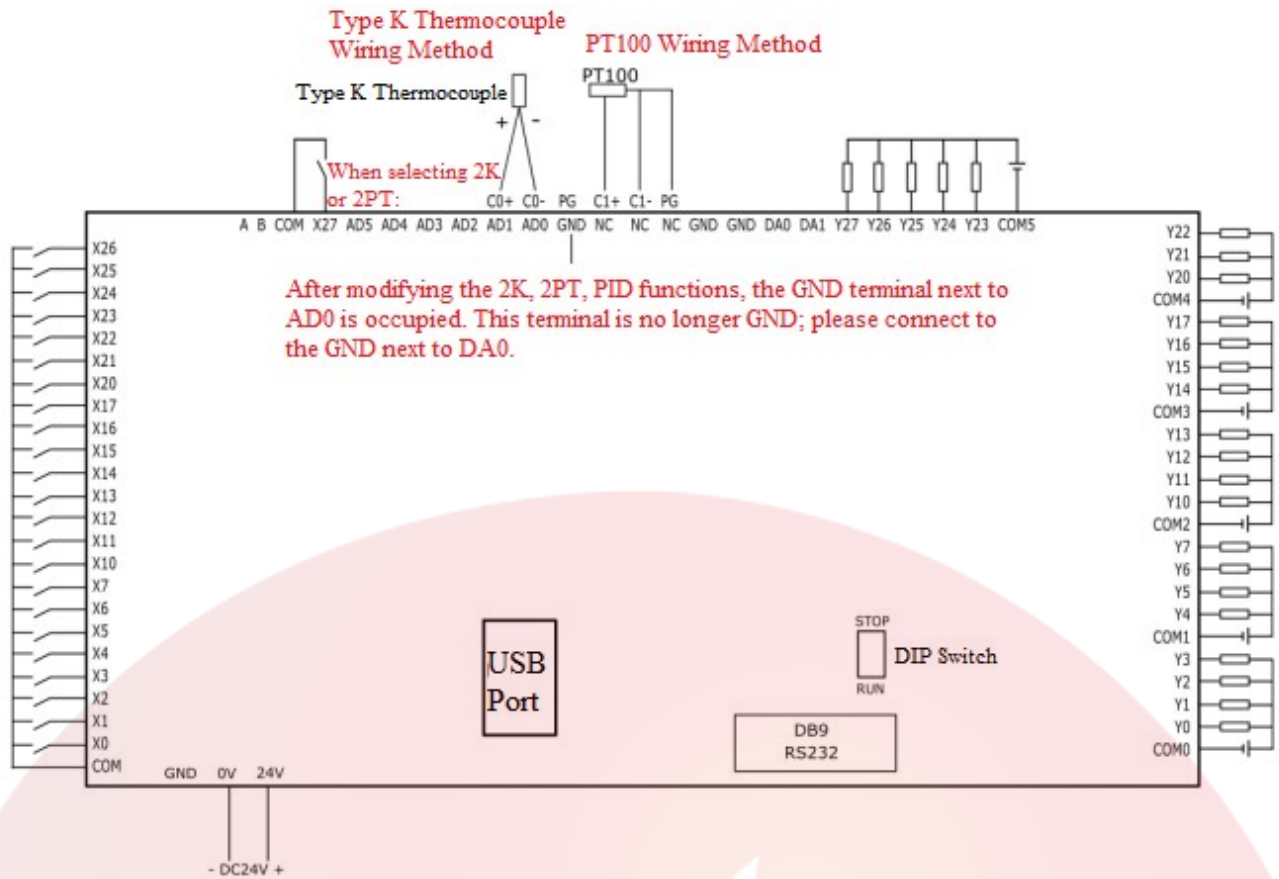


Figure 4-15

Chapter 5 PLC Partial Programming Reference

5.1 Application environment

1. GX Developer (compatible with XP and 32-bit Windows 7 systems)
2. GX Works2 (compatible with 64-bit Windows 7, Windows 8, and Windows 10 systems)

5.2 Soft Component Function Overview

Auxiliary relay M	M0-M3071 (with M500-M1023 supporting power loss preservation range)
Step point S	S0-1023 (where S500-S1023 denotes the power loss preservation range)
100ms timer	T0-T199 (where T184-T199 represents the cumulative power loss range)
10ms Timer	T200-T249 (where T246-T249 represents the cumulative power loss range)
1Ms Timer	T250-T383 (where T250-T255 represents the cumulative power failure range)
16-bit counter	C0-C199 (where C100-C199 are the supported power loss recovery ranges)
32-bit counter	C200-C234 (with C220-C234 supporting power loss recovery range)
32-bit high-speed counter	C235-255 and C235-240 are single-phase counters without frequency doubling; C241-240 is a single-phase counter with 2x frequency doubling; C247-249 is a two-phase counter without frequency doubling; C250-252 is a two-phase counter with 2x frequency doubling; C253-255 is a two-phase counter with 4x frequency doubling.
Register D	D0-D7999 (where D200-D7999 supports power loss recovery range)
Indirectly addressed pointers V and Z	V0-7, Z0-7
P subroutine jump number	P0-63
Interrupt	X0-5 External interrupt. Timer interrupt (1MS units). Counter interrupt.
Special M Element	The M8000 operates with continuous operation mode, the M8002 uses power-on pulses, the M8011 delivers 10ms pulses, the M8012 provides 100ms pulses, the M8013 generates 1-second pulses, and the M8014 outputs 1-minute pulses.

Table 5-1 Summary of Soft Component Functions

5.3 Basic Instructions

No.	Mnemonic	Function	No.	Mnemonic	Function
1	LD	The operation starts with Normally Open Contact	17	RST	Releasing coil action holding
2	LDI	The operation starts with Normally Closed Contact	18	PLS	Coil rising edge output
3	LDP	Rising edge detecting with operation beginning.	19	PLF	Coil falling edge output
4	LDF	Falling edge detecting with operation beginning.	20	ALT	Alternate output
5	AND	Normally Open Contact in series	21	MC	Common string contact with coil command
6	ANI	Normally Closed Contact in series	22	MCR	The command of releasing common contact
7	ANDP	Rising edge detecting in Series connection	23	MPS	Operation storage
8	ANDF	Falling edge detecting in Series connection	24	MRD	Storage reading
9	OR	Normally Open Contact in parallel	25	MPP	Storage reading and resetting
10	ORI	Normally Closed Contact in parallel	26	INV	Negation of operation result
11	ORP	Rising edge detecting in parallel connection	27	END	The end of program
12	ORF	Falling edge detecting in parallel connection	28	STL	Step ladder start
13	ANB	Circuit blocks are connected in series.	29	RET	The end of step ladder
14	ORB	Circuit blocks are connected parallel		CALL	Subroutine call
15	OUT	Coil output driving		SRET	Subroutine return
16	SET	Coil action holding			

Table 5-2 Basic Instructions

5.4 Application Instruction

Sort	No.	Mnemonic	Function
Program flow	1	CJ	Conditional Jump
	2	CALL	Subroutine call
	3	SRET	Subroutine Return
	4	FEND	End of main program
	5	FOR	Loop Range start
	6	NEXT	Loop Range end
Transmission and comparison	7	CMP	Comparison
	8	ZCP	Regional comparison
	9	MOV	Transmission
	10	CML	Reverse transmission
	11	BMOV	Simultaneous transmission
	12	FMOV	multicast
	13	XCH	Exchange
	14	BCD	BCD conversion
	15	BIN	BIN conversion
Four logical operations	16	ADD	BIN Addition
	17	SUB	BIN Subtraction
	18	MUL	BIN Multiplication
	19	DIV	BIN Division
	20	INC	BIN Add 1
	21	DEC	BIN Minus 1
	22	WAND	Logical word and
	23	WPR	Logical word or
	24	WXOR	Logical Exclusive OR
	25	NEG	Complement code
Cyclic shift	26	ROR	Ring shift right
	27	ROL	Ring shift left
	28	RCR	Right shift
	29	RCL	Left shift
	30	SFTL	Bit left shift
	31	SFTR	Bit Shift Right
Bit data	32	ZRST	Batch Reset

	33	MEAN	Average
	34	FLT	BIN integer→Binary floating-point conversion
	35	GRY	BIN integer→Gray Code Conversion
	36	GBIN	Gray code →BIN integer
High Speed Processing	37	DHSCS	High Speed Comparison setting
	38	DHSCR	High Speed Comparison Resetting
	39	SPD	Pulse density and pulse width (pulse interval time) can also be measured.
	40	PLSY	Pulse Output
	41	PLSV	Pulse output with direction control
	42	PWM	Pulse Width Modulation, 0-32767us
	43	PLSR	Pulse output with acceleration and deceleration
	44	DRVA	Absolute position control
	45	DRVI	Relative position control
	46	ABSD	Cam Control (Absolute Mode)
Peripheral Equipment	47	RS	Serial data transfer
	48	ASCI	HEX - ASCII conversion
	49	HEX	ASCII - HEX conversion
	50	CCD	Check code
	51	PID	PID operation
	52	SEGD	BCD to 7-segment code digital tube
Floating-point Arithmetic	53	ECMP	The comparison of binary floating point number
	54	EZCP	Interval comparison of binary floating point number
	55	EBIN	The conversion between decimal floating point number and binary floating point number
	56	EADD	Addition of binary floating point number
	57	ESUB	Subtraction of binary floating point number
	58	EMUL	Multiplication of binary floating point number
	59	EDIV	Division of binary floating point number
	60	INT	Conversion between binary floating point number and BIN integer
	61	SIN	Operation of floating point number SIN
	62	TAN	Operation of floating point number TAN
	63	COS	Operation of floating point number COS
	64	ASIN	Operation of floating point number SIN-1
	65	ATAN	Operation of floating point number TAN-1
	66	ACOS	Operation of floating point number COS-1

	67	EXP	Exponent arithmetic of binary floating point number
	68	LOGE	Natural Logarithm Operation of Binary Floating Point Number
	69	LOGE10	Common Logarithmic Operation of Binary Floating Point
	70	SWAP	Conversion between Up and down byte
Convenient Command	71	SER	Data lookup
	72	ALT	Alternating output
	73	RAMP	Ramp signal
	74	BON	ON bit Decision
	75	SUM	ON bits
	76	ANS	Alarm set
	77	ANR	Alarm Reset
	78	HOUR	Chronograph
Clock Instruction	79	TRD	Clock data reading
	80	TWR	Clock data writing
Contact Comparison	81	LD=	(S1)=(S2)
	82	LD>	(S1)>(S2)
	83	LD<	(S1)<(S2)
	84	LD◇	(S1)≠(S2)
	85	LD≤	(S1)≤(S2)
	86	LD≥	(S1)≥(S2)
	87	AND=	(S1)=(S2)
	88	AND>	(S1)>(S2)
	89	AND<	(S1)<(S2)
	90	AND◇	(S1)≠(S2)
	91	AND≤	(S1)≤(S2)
	92	AND≥	(S1)≥(S2)
	93	OR=	(S1)=(S2)
	94	OR>	(S1)>(S2)
	95	OR<	(S1)<(S2)
	96	OR◇	(S1)≠(S2)
	97	OR≤	(S1)≤(S2)
	98	OR≥	(S1)≥(S2)

Table 5-3 Application Instructions

Note: Supports 32-bit instructions and pulse-execution instructions

5.5 This version does not support the command list.

No.	Mnemonic	Function
1	ZRN	Regression Through the Origin, only 16-bit command supported
2	DSZR	Regression Through the Origin with DOG search
3	DVIT	Interrupt positioning

Chapter 6 Common Issues and Solutions

No.	Questions/Problem	Solutions
1	Wiring method of Analog	The signal line is connected to the corresponding analog ports (AD0–AD2), and the negative terminal is connected to GND.
2	Analog reading	Refer to Section 4.5
3	Encryption	When online communication is confirmed: 1.Set the button switch to STOP while the indicator light flashes; 2.Enter the "keyword"; 3. After setting the password twice, return the button switch to its original position.
4	Download cable wiring sequence	1. For touchscreen download: USB male-to-male download cable 2. For PLC download: Standard 232 port with cable sequence 2-2, 3-3, 5-5; if the computer has a USB interface, add a USB-to-232 serial cable
5	What is the product's power consumption?	Refer to Section 4.1
6	The product cannot communicate with the computer.	1.Check whether the communication cables are properly connected and the drivers are installed; 2. Verify that the PLC series, baud rate, and COM port settings in the software are correct.
7	How to connect the common input terminal	All input terminals are NPN inputs, with the negative terminal conducting.
8	The PLC cannot communicate with the touchscreen.	After the program download is complete, disconnect the download cable; the PLC and touchscreen will automatically communicate (Note: When the download cable is connected, the PLC and touchscreen cannot communicate).

Chapter 7 Warranty Terms

7.1 Warranty period: 12 months

The product comes with a one-year warranty from the date of shipment. During the warranty period, our company provides free repair services for the product.

7.2 Not covered by warranty

- Improper wiring, such as reversing the positive and negative power supply terminals.
 - Use beyond the voltage range or environmental requirements.
 - Unauthorized modification of internal components.
- 